

ADAM LARSON

SCIENTIFIC SOFTWARE
MODELING & SIMULATION

Contact

San Jose, California

admlars2@gmail.com

Open to relocation and remote,
hybrid, or on-site work

Links

[Portfolio](#)

[GitHub](#)

[LinkedIn](#)

Skills

PROGRAMMING

Python
TypeScript / JavaScript
C++ | C# | SQL

MODELING & VISUALIZATION

Three.js | WebGL | React
Procedural modeling
Deterministic systems
Data analysis

DATA & SYSTEMS

OpenPyXL | FastAPI
MongoDB | Redis
MQTT | Azure IoT Hub
Git | Linux | Docker

MEASUREMENT

Metrology
Blueprint reading
CMM Manager
Precision measurement

Education

San Diego State University

B.S. Computer Science
2025

COURSEWORK

Machine Learning
Systems Programming
Operating Systems
Database Theory
Cloud Computing

Professional Summary

Computer science graduate building deterministic modeling, data, and visualization tools for physical systems. Experience spans procedural planet generation, backend and data systems, and precision manufacturing measurement. Seeking scientific or research software, Earth/geospatial data, and modeling or simulation roles.

Experience

Inspector / Programmer

PHOENIX PRECISION MACHINING | SANTA CLARA, CA | JUN 2026 - PRESENT

- Inspect precision-machined components against engineering drawings using micrometers, calipers, bore micrometers, pin gauges, CMM Manager, and coordinate-measuring equipment.
- Read blueprints and verify dimensions against tolerances regularly specified in thousandths and ten-thousandths of an inch across CNC-, five-axis-, and EDM-produced parts.
- Develop an internal Python/OpenPyXL tool for inspection-data aggregation.

Selected Projects

Tectonic Planet Generator | *TypeScript, React, Three.js/WebGL, Web Workers*

LIVE DEMO | GITHUB

- Built and deployed a deterministic, tectonics-inspired procedural planet generator on approximate spherical neighbor graphs; seeded stages produce connected plate regions, heuristic boundaries, terrain fields, targeted ocean coverage, and exportable textures and 16-bit height maps.
- Structured generation as a worker-backed pipeline with progress, cancellation, caching, typed stage contracts, fixed-seed regression tests, and stage-level semantic hashes; exposed intermediate behavior and documented the model's scientific limits.

IoT Sensor Simulator | *Python, MQTT, Azure IoT Hub*

- Developed Python scripts that generated sensor readings and transmitted JSON messages through MQTT for a team-built monitoring workflow with dashboard visualization and threshold alerts.

Spaceship Titanic ML Pipeline | *Python, CatBoost, HistGradientBoosting*

- Built a leakage-safe GroupKFold pipeline with shared train/test feature engineering; achieved a 0.804 public Kaggle score and rank 627 at submission.

LangLift | *Python, FastAPI, MongoDB, Redis* | **Case study**

- Designed MongoDB domain models and async data-access layers for cards, decks, study sets, users, daily posts, attachments, metrics, and per-user review history.
- Built a service-oriented FastAPI backend with separated API, workflow, and database layers plus Redis-backed caching, real-time connection state, background workflows, and service health checks.